

Supplementary Materials for Matrix-Free Covariance Scaling for Recorded Ground-Motion Signals

Molena Huynh

1 Complete rejection-capable algorithm

Let $A(t)$ be a target action available at selected task values and let $\mathcal{B}_r(t)$ be a sum of r butterfly products. The implementation executes the following fixed protocol. Training and certification random-number generators are initialized from separate seeds; no certification vector appears in optimization.

The fixed protocol is:

1. Fix five optimization seeds, candidate grid $\{1, 2, 4\}$, five certified tasks, $\tau = 0.15$, global $\delta = 0.05$, and 2,048 probes. The complete family has $J = 75$ hypotheses.
2. Fit all candidates with 1,500 minibatches of 48 action pairs and learning rate 0.006; freeze each model before certification.
3. At every task, draw the registered Gaussian block and evaluate residual and target actions. Use $t = \log(2J/\delta)$ in the simultaneous certificate from the main manuscript.
4. For each seed, accept the first candidate whose maximum five-task upper bound is at most 0.15. If none passes, return rejection. Entry-built controls and the apply/adjoint structural diagnostic cannot change this decision.

For one product at dimension $n = 2^L$, the forward recurrence stores the stage inputs

$$z_{\ell+1} = F_\ell(t)z_\ell, \quad \ell = 0, \dots, L-1.$$

Given residual $e = z_L - y$, the reverse recurrence

$$a_L = e, \quad a_\ell = F_\ell(t)^\top a_{\ell+1}$$

gives the gradient of every active 2×2 core as the corresponding outer product of entries of $a_{\ell+1}$ and z_ℓ . The implementation performs these indexed core products directly and never differentiates an assembled dense target. For one right-hand side, forward application, adjoint application, and a gradient step each use $O(rn \log n)$ arithmetic. The untied factors store $O(rn \log n)$ scalars; the released reverse pass also caches $O(rn \log n)$ activation scalars. A streaming action alone needs only $O(n)$ working storage beyond the factors.

2 Frozen source and split

The primary experiment contains 48 EarthScope requests: three stations times four earthquakes times one event and one quiet window in each of the training and test partitions. Each request asks for three broadband components through the BH? selector. Event windows begin 900 seconds after the USGS origin; quiet windows use the same clock time 24 hours earlier. No waveform is redistributed in the submission package.

Table S1: Frozen earthquake and station split. Event identifiers are USGS ComCat identifiers; stations are IU Global Seismographic Network codes.

Partition	Event identifier	Year	Stations
Training	official20100227063411530_30	2010	ANMO, COLA, MAJO
Training	official20110311054624120_30	2011	ANMO, COLA, MAJO
Training	official20120411083836720_20	2012	ANMO, COLA, MAJO
Training	usb000h4jh	2013	ANMO, COLA, MAJO
Test	usc000nzvd	2014	COR, KONO, HRV
Test	us20003k7a	2015	COR, KONO, HRV
Test	us2000ahv0	2017	COR, KONO, HRV
Test	us60003sc0	2019	COR, KONO, HRV

The parser handles segmented GeoCSV responses, chooses two horizontal components and one vertical component, and interpolates them to the common 20-Hz time grid. A request is rejected if coverage is below 95% or any gap exceeds one second. The accepted requests supply either 4,799 or 4,800 common samples; the largest observed gap is approximately 0.05 seconds. The source records the URL, instrument string, units, sample rate, byte length, selected channels, and SHA-256 digest for each request. The 48 payloads total 29,029,069 bytes.

The preprocessing removes the mean from each 64-sample component window, concatenates the three components, zero pads the 192 samples to $n = 256$, and normalizes each nonzero window to unit Euclidean norm. No taper or featurewise standardization is applied. EarthScope `scale=AUTO` supplies velocity units through the service-reported scalar sensitivity. This is not a claim of full poles-and-zeros response deconvolution.

3 Additional locked numerical details

Table S2: Worst five-task certificate for the four-product candidate. In all seeds the worst task is $t = -4$; every upper bound exceeds 0.15.

Training seed	Certificate seed	Relative estimate	Upper bound
260726	1000020	0.139921	0.181167
260727	1000120	0.136690	0.176984
260728	1000220	0.153743	0.199064
260729	1000320	0.144100	0.186578
260730	1000420	0.143688	0.186045

Table S3: Apply/adjoint-probe structural diagnostic at relative residual tolerance 0.05. These sampled values are not rank upper bounds or a gate.

Duration (s)	Dimension	Maximum tested rank	Saturated blocks
0.8	64	8	0
1.6	128	8	0
3.2	256	14	0
6.4	512	16	0

The action-probe audit does not prove asymptotic compressibility and is not used for deployment.

4 Reproduction and falsification

```
python -m venv .venv
.venv/bin/python -m pip install --upgrade pip
.venv/bin/python -m pip install "./code[test]"
.venv/bin/python -m pytest code/tests
.venv/bin/taskbutterfly-reproduce --verify > reproduced.json
```

The verifier rechecks every source digest and deterministic scientific quantity. It intentionally excludes wall-clock timing and BLAS-sensitive measurements. A candidate is rejected if any fixed task's simultaneous upper bound exceeds 0.15; this occurs for all candidates in the registered replay. The seismic endpoint does not support a claim of operational event detection, phase picking, or superiority over raw amplitude.